

- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5	✓	✓		✓
CO 6	✓			

Programme	BSc Statistics				
Course Code	STA3MN210 (P)				
Course Title	Probability theory and sampling techniques				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Understanding of basic algebraic operations and set theory. Familiarity with functions, graphs and their properties.				
Course Summary	Through theoretical concepts and practical applications, students will develop the skills necessary to analyze uncertainty, conduct sample surveys, and implement statistical quality control methods.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and apply basic concepts of probability, including classical probability and the axiomatic approach, to analyze uncertain events and outcomes.	U	C	Instructor-created exams / Quiz
CO2	Differentiate between census and sampling methods and recognize the advantages and limitations of each approach in data collection and critically evaluate ethical implications of statistical methods aligning with human values.	Ap	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Describe the principal steps involved in sample surveys, including the organization and execution of large sample surveys.	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Define life tables and understand their construction, including calculating measures such as the force of mortality, and interpreting results in demographic analysis and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Instructor-created exams / Home Assignments
CO5	Explain the concept of statistical quality control and differentiate between assignable causes and chance causes of variation in processes.	Ap	C	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Create basic plots in R to visualize range, variance and correlation between variables.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Probability		10	15
	1	Basic concepts of Probability ,Classical definition of Probability ,Axiomatic approach to Probability	2	
	2	Addition Theorem, Multiplication Theorem	3	
	3	Conditional Probability	3	
	4	Independence of events	2	
	Sections from References: Unit 1: 3.3-3.8 [Ref 1] Unit 2: 3.9,3.11 [Ref 1] Unit 3: 3.10[Ref 1] Unit 4: 3.11-3.15 [Ref 1]			
II	Random Variables		8	15
	5	Random Variables, Discrete and continuous random variables	2	
	6	Probability distribution , Distribution function (Applications in discrete case)	2	
	7	Mathematical expectation (Applications in discrete case)	2	
	8	Variance (Applications in discrete case)	2	
	Sections from References: Unit 5&6: 5.1-5.4.2 [Ref 1] Unit 7: 6.1-6.4 [Ref 1] Unit 8: 6.6 [Ref 1]			
III	Sampling theory		19	25
	9	Population and Sample	2	
	10	Census and Sampling Method	3	
	11	Advantages and Limitations of Sampling	1	
	12	principal steps in sample survey	3	
	13	Sampling Errors	3	
	14	Non-Sampling Errors	3	
	15	Simple random sampling(Concept and Methods of selection)	1	
	16	Stratified random sampling	1	
	17	Systematic Sampling	2	
	Sections from References: Unit 9: 15.2,15.3,15.6 [Ref 1] Unit 10: 15.6,15.7[Ref 1] Unit 11: 15.8 [Ref 1] Unit 12: 15.9.1[Ref 1] Unit 13&14:15.10[Ref 1] Unit 15:15.11,15.11.1 [Ref 1] Unit 17: 15.12,15.12.1 [Ref 1] Unit 17: 15.13 [Ref 1]			

IV	R programming		8	15
	18	Range	2	
	19	Variance	2	
	20	Loops- Brief explanation	2	
	21	Pearson’s correlation	1	
	22	Conditional statements(Brief)	1	
	Sections from References: Unit 18&19: 2.1-2.3[Ref 3] Unit 20: 7.5 [Ref 2] Unit 21: 6.2 [Ref 2] Unit 22: 7.3 [Ref 2]			
V	PRACTICUM		30	
		Do practice problems in R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course. 1. Range 2. Mean Deviation 3. Quartile Deviation 4. Standard Deviation 5. Variance 6. Covariance 7. Correlation 8. Rank correlation		
	Sections from References: Unit 1,2: 2.1 [Ref 3] Unit 3: 2.2 [Ref 3] Unit 4,5: 2.3 [Ref 3] Unit 6,7,8: 6.2-7.5 [Ref 2]			
Books and References: 1. Gupta, S. C. and Kapoor, V. K. (2020). Fundamentals of Mathematical Statistics, 12 th edition, Sulthan Chand, New Delhi 2. Douglas, Alex, Deon Roos, Francesca Mancini, Ana Couto, and David Lusseau. (2020), An Introduction to R. https://intro2r.com/index.html . 3. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh (2015), Statistics Using R.				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
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CO 1	2	1	-	-	-	-	3	-	-	-	-	-
CO 2	-	-	1	-	-	2	-	-	-	-	-	3
CO 3	-	-	-	1	2	-	-	-	3	-	2	-
CO 4	-	-	-	-	-	2	-	1	2	-	-	-
CO 5	-	-	2	-	-	-	1	-	-	-	-	-
CO 6	-	-	3	-	-	2	-	-	-	2	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
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CO 1	✓			✓
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